

Tianyu Kong

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Research Interests

Mathematical modeling, numerical analysis, partial differential equations, multiple-scale analysis, mathematical physics, quantum physics, and optimization

Education

University of Minnesota Twin Cities, Minneapolis, MN

09.2021 – 2026 (Expected)

PhD in Applied Mathematics

- Advisor: Mitchell Luskin, Alexander B. Watson
- GPA: 3.96/4.00

University of Chicago, Chicago, IL

09.2017 – 06.2021

Honors BS in Applied Mathematics & Physics

- GPA: 3.73/4.00

Honors and Scholarships

SIAM Student Travel Award

03.2024

Vanky Men Memorial Fellowship (UMN)

12.2023

Dean's List (UChicago)

07.2018, 07.2019

Publications

1. **Modeling and computing many-body electronic properties of twisted bilayer graphene with mechanical relaxation**
Tianyu Kong, Alexander B. Watson, Mitchell Luskin, Kevin D. Stubbs
arXiv preprint. arXiv:2504.03479
2. **Higher-order continuum models for twisted bilayer graphene**
Solomon Quinn, Tianyu Kong, Mitchell Luskin, Alexander B. Watson
arXiv preprint. arXiv:2502.08120
3. **Modeling of electronic dynamics in twisted bilayer graphene**
Tianyu Kong, Diyi Liu, Mitchell Luskin, Alexander B. Watson
SIAM Journal on Applied Mathematics. 84, 1011 (2024).
4. **Bistritzer-MacDonald dynamics in twisted bilayer graphene**
Alexander B. Watson, Tianyu Kong, Allan H. MacDonald, Mitchell Luskin
Journal of Mathematical Physics 64.3 (2023). Editor's Choice

Research Experience

Graduate Research Assistant, UMN

06.2021 – Present

Advisor: Mitchell Luskin and Alexander B. Watson

- Studied mathematical modeling and numerical analysis for incommensurate 2D materials, specifically twisted bilayer graphene
- Accurately computed the band structure and single electron dynamics in tight binding models of such systems
- Used multiple-scale analysis to identify a regime where the discrete aperiodic model can be reduced to a periodic continuum model

- Derived a domain truncation scheme to approximate the electron dynamics in an infinite system.
- Implemented Python code to numerically verify results

Student Assistant, Lawrence Berkeley National Laboratory

05.2024 – 08.2024

Advisor: Chao Yang and Lin Lin

05.2023 – 08.2023

- Studied exotic quantum phases in magic angle twisted bilayer graphene using interacting models that accounts for electron-electron correlation
- Implemented Hartree-Fock (HF) and Coupled Clusters (CC) methods with Python-based Simulation of Chemistry Framework (PySCF)
- Developed an accurate continuum model to compute the electron dispersion with structural relaxation, and computed long range Coulomb interaction between electrons

Undergraduate Research Assistant, UChicago

04.2020 – 05.2021

Advisor: Mary Silber

- Studied different types of Minimum Action Methods to determine the transition of states in a dynamical system subject to random perturbation
- Implemented MATLAB code and used steepest descent and quasi-Newton methods to calculate the Minimum Action Path in Lorenz systems

Teaching Experience

Teaching assistant, School of Mathematics, UMN

09.2021 – 05.2024

Honors Calculus

Calculus for College of Science and Engineering

- Held 2 workshop sessions per week for 3 hours total, lectured and organized group discussions on challenging concepts and problems, and responded to students' questions in an engaging manner.
- Graded weekly homework, quizzes and final exams promptly, and provided valuable feedback.

Grader, School of Mathematics, UMN

09.2022 – 05.2024

Mathematical Modeling and Applied Mathematics

Functional Analysis

- Promptly graded homework and exams for graduate level courses.

Grader, Department of Mathematics, UChicago

09.2018 – 05.2020

Mathematical Methods for Physical Sciences

Honors Calculus

- Promptly graded weekly homework and exams for ~ 30 students.

Talks and Seminar Presentations

Modeling Electron Interactions in Twisted Bilayer Graphene

03.2025

UMN Quantum Math Seminar, Minneapolis MN

“Magic” in Moiré Materials – An Applied Mathematics Perspective

03.2025

UMN Seminar of Mathematical Applications & Computations, Minneapolis MN

Modeling of Electronic Dynamics in Twisted Bilayer Graphene

07.2024

International workshop on 2D and moiré materials, Roscoff France

Modeling of Electronic Dynamics in Twisted Bilayer Graphene

05.2024

SIAM Conference on Materials Science (MS24), Pittsburgh, PA

Modeling of Electronic Dynamics in Twisted Bilayer Graphene

03.2024

Brin Mathematics Research Center Workshop, College Park, MD

Poster Presentations

A Comparison of Minimum Action Methods for Computing Noise-induced Transitions of the Lorenz System

05.2021

SIAM Conference on Applications of Dynamical Systems (DS21), Remote

Professional Affiliations

Webmaster, SIAM student chapter, UMN

09.2024 – Present

- Managed website and documents for SIAM student chapter at UMN
- Co-organized annual student Integration Bee contest

Technologies

Python, MATLAB, Julia, R, MySQL